Two-Regime Models for Daily Soil-Moisture Estimation in Washington

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SUMMER RESEARCH SHOWCASE · 2026

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Background & Research Question

Why soil moisture estimation?

Soil moisture describes water stored near the land surface. It informs irrigation, drought monitoring, and our understanding of water availability. Estimates help us follow rainfall responses and drying where direct measurements are sparse.

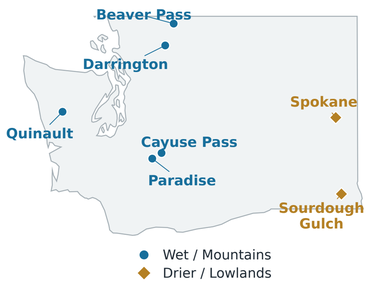
Ground sensors measure local conditions; satellites offer broader coverage, but leave gaps between observations and missing retrievals. Our project combines satellite, weather, and site information for daily estimates. Earlier work established this pipeline and a single global XGBoost predictor.

THE SUMMER RESEARCH QUESTION

Can two specialists improve estimates across Washington's contrasting conditions?

This summer's work tests feature-based specialization against the global predictor, adds consistent assignment for known stations, and examines how missing routing inputs affect estimates at new sensor sites.

Study Area & Data



Regional contrast

Washington spans coastal sites, Cascade mountains, and a drier east. We ask whether two specialists can learn distinct soil responses in these settings. Each station holdout tests how well these patterns carry to a new site.

Figure 1. Seven Washington training stations and their fitted memberships; points do not define geographic regime boundaries.

NOAA USCRN and SNOTEL supply in-situ observations. Inputs combine SMAP moisture, Sentinel radar and optical measurements, MODIS temperature and vegetation, ERA5-Land weather, and terrain, climate, land-cover, and soil descriptors.

The pipeline aligns sources daily, fills observation gaps, smooths the series, and derives recent-history features. Both approaches receive identical prepared inputs, keeping the comparison focused on grouping and assignment.

2017–2022
Final fitting period → **2023–2025**
Later-year test

Development uses 2017–2020 training and 2021–2022 validation data; the final fit combines these periods. Evaluation tests later years; leave-one-station-out evaluation refits without each station to test new Washington locations.

Two-Regime Modeling

A global model fits all observations together. Our model groups observations by their inputs, then fits an XGBoost specialist per group. The target is daily volumetric soil moisture at 5 cm depth (m^3/m^3).

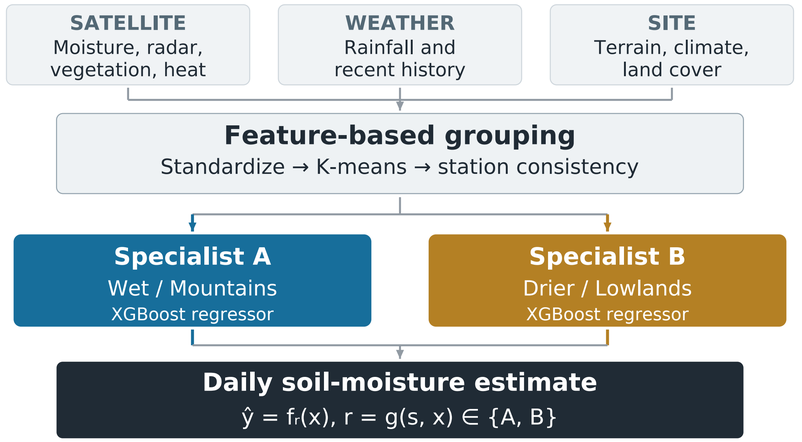


Figure 2. Hard routing. x denotes inputs, s the station, g the assignment rule, and f_r the selected specialist; one observation produces one estimate.

1. Learn groups. Fill missing router values with fitting-data means and standardize features before K-means finds two centroids. New observations use the nearest centroid. Clustering uses no observed in-situ targets.

2. Keep stations consistent. Known stations receive their majority fitting-data group, keeping observations with one specialist. New stations use row-level assignment; availability checks can trigger the fallback described at right.

3. Isolate specialization. Both approaches use identical features, XGBoost settings, evaluation rows, and matched seeds. Specialists learn only from assigned observations, so differences reflect grouping and assignment under this protocol.

Regime Interpretation

Wet / Mountains
5 fitted stations
Western and Cascade sites; more rain and higher satellite moisture.
Wettest-month rain: 391 mm
Satellite moisture: 0.414 m^3/m^3

Drier / Lowlands
2 fitted stations
Spokane and Sourdough Gulch; less rain and lower satellite moisture.
Wettest-month rain: 70 mm
Satellite moisture: 0.192 m^3/m^3

These names describe this Washington sample; elevation alone does not determine membership. The wet group includes lowland and mountain stations. Medians summarize input conditions, not measured-moisture thresholds.

Specialists can learn narrower relationships, but require reliable assignment and representative training data. Select regime counts using local data; two groups are useful here.

Evaluation & Results

+0.0320
Later-year R^2 gain

+0.0581
Station-holdout R^2 gain

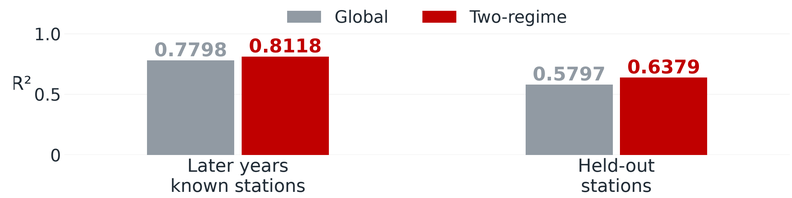


Figure 3. Mean R^2: 30 matched temporal seeds; five per station holdout. Higher is better.

Two-regime R^2 leads in all 30 later-year runs. Station-holdout performance improves on average, with uneven benefits across locations.

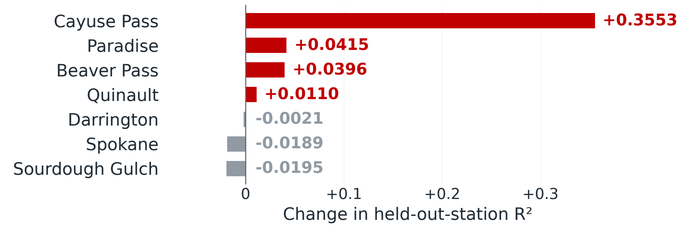


Figure 4. Five-seed station differences: four gains and three losses.

Cayuse Pass gains most. Eastern holdouts reduce representation of the smaller regime. Seeds describe fitting variability, excluding uncertainty across new sites and seasons.

ECE Sensor-Site Evaluation

Five new sites deployed by Seattle University's ECE team lacked all SMAP inputs. An automatic availability gate switched to antecedent-precipitation routing, with the specialist mapping selected using Washington validation.

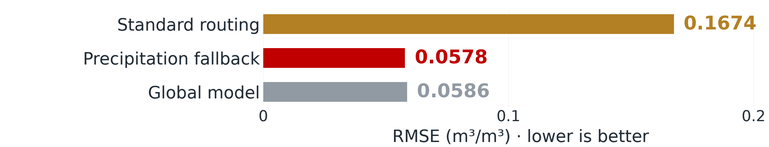


Figure 5. ECE pooled RMSE: fallback approaches global-model error. Lower is better.

Across 150 daily observations (July 20–August 19, 2026), every fallback row used one specialist. This short summer test supports a routing lesson, without establishing full-season superiority.

Discussion & Next Steps

Specialization helps with reliable assignment and representative training sites. Next, evaluate more Washington stations and seasons with ECE sensors, and test routing when satellite coverage changes.